

Physics-Constrained Rectified Flow for Climate Downscaling

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Abstract

Statistical downscaling maps a coarse climate projection or reanalysis to a high-resolution regional field, restoring the mesoscale structure behind flooding and crop stress. Diffusion and flow-matching generators can produce sharp fields that still violate the conservation laws in their coarse conditioning. We present Physics-Constrained Rectified Flow (PC-RF), a conditional flow-matching downscaler constrained in training and at inference. Three differentiable penalties on a one-step clean-field estimate target wind divergence, negative precipitation, and mass drift from the coarse integral, and an inference projection makes the last two constraints exact. On ERA5 our divergence operator took the northward derivative with the wrong sign, so the ERA5 divergence penalty and Divergence Error (DE) acted on a deformation-like term. On one synthetic field where no network is trained, the projection drives the Non-negativity Violation Rate (NVR) of a corrupted baseline field from 0.236 to zero. It also drives the field’s Mass-Conservation Error (MCE) from 0.184 to zero, and the penalties lower its DE by about 30%. On ERA5 2020 the projection gives PC-RF and the unconstrained RF-base an NVR and MCE of exactly zero. PC-RF scores slightly above RF-base (SSIM 0.914 vs. 0.912, PSNR 26.59 vs. 26.39, CRPS 1.149 vs. 1.188), but the two runs had different training budgets. In a matched-budget ablation the SSIM gap closes and a smaller PSNR and CRPS lead remains, which a single training run per arm cannot separate from run-to-run variation. We also compare bicubic interpolation and four reimplemented baselines, all scored without the projection. CorrDiff beats both flow models on PSNR (28.38) and CRPS (1.092), yet 45% of the cells in its ensemble-mean precipitation are negative. A three-year rerun on 2019–2021 at a matched budget shows no perceptual edge for PC-RF (SSIM 0.939 vs. 0.942, PSNR 31.81 vs. 31.79, CRPS 0.771 vs. 0.772 for RF-base), while NVR and MCE stay exactly zero for both, so the projection provides the guarantee. Our fine reference is interpolated from ERA5 and accumulates precipitation over a longer window than the coarse input, so the perceptual scores on real data cannot establish downscaling skill. The validity metrics never use the reference. NVR and MCE serve as pass-or-fail checks, and a correctly computed DE serves as a diagnostic. All three need one pass over the output and no high-resolution reference, so we argue they should run first when verifying generative climate fields.

1 Introduction

Statistical downscaling estimates a high-resolution regional field from a coarse one and puts back the sub-grid texture of a variable such as precipitation or near-surface wind. Adaptation planning depends on that detail, because the harms of a warming climate arrive at local scales that a coarse grid does not resolve. We work with reanalysis, where ERA5 (Hersbach et al., 2020) arrives on a 0.25° grid, too coarse for the local variability behind flooding and crop stress.

Two properties make the task hard. A climate field ties spatial and temporal dependence together under several drivers at once, and it is non-stationary, so a model fit to one stretch of years carries no promise of holding in the next. Recovering a fine field from a coarse one is also ill-posed, since nonlinear coupling and model uncertainty leave the answer non-unique. The task calls for a method that is both physically grounded and probabilistic.

The reference itself adds a further difficulty. A reanalysis such as ERA5 is a model product that assimilates partial observations into a global record, so the fine field a downscaler is scored against carries errors of its own, and any perceptual score taken against it inherits them. The conservation laws

a field must obey give a second check that asks for no reference at all. Non-negative precipitation and a domain-mean precipitation that matches the coarse input can be tested on the generated field and its coarse input alone. A hard projection can make both hold by construction. We therefore distinguish what a soft training penalty is meant to shape from what a hard inference projection certifies.

Deep generative models now downscale and forecast weather and climate fields. CorrDiff (Mardani et al., 2025) uses diffusion to downscale a 25 km reanalysis to 2 km wind, temperature, and a radar-reflectivity surrogate for rain over Taiwan. GenCast (Price et al., 2025) uses diffusion for global ensemble forecasts at 0.25° , and data-driven forecasters such as GraphCast (Lam et al., 2023) have reshaped medium-range prediction. Rectified flow (Liu et al., 2023; Lipman et al., 2023) straightens the generative path from noise to data and samples in far fewer network evaluations, which matters at downscaling resolutions. A natural-image super-resolution network borrowed wholesale imposes no physics, and a pixel-wise or perceptual objective gives no guarantee that the output respects the atmospheric physics carried in its coarse conditioning. Generated wind can diverge, precipitation can go negative, and the domain mass can drift from the coarse input.

Physics-informed neural networks (Raissi et al., 2019) embed PDE residuals as soft constraints. Other work builds the physics into the model, as a learned solver that keeps the flow incompressible by construction (Kochkov et al., 2021) or as normalizing flows that respect molecular symmetries exactly (Köhler et al., 2020). For downscaling, constraint layers can enforce conservation exactly at the output of a network (Harder et al., 2023). Recent work adds soft per-cell conservation penalties for precipitation and humidity to an adaptive flow-matching downscaler (Debeire et al., 2026). Hard constraints at sampling time already exist for pretrained flow-matching models of PDE systems (Cheng et al., 2025; Utkarsh et al., 2025) and for a consistency-model downscaler (Bochow et al., 2026). Hess et al. (2022) add one global rescaling of precipitation to a trained generative adversarial network. PC-RF applies the same domain-level rescaling, after a rectifier, to the precipitation samples of a rectified-flow downscaler. Training penalties are meant to shape the generated field, and the inference-time projection makes its non-negativity and domain mass balance exact.

Contributions: We make three contributions.

- We propose Physics-Constrained Rectified Flow (PC-RF), a conditional rectified-flow downscaler whose three differentiable training penalties target wind divergence, negative precipitation, and mass drift from the coarse integral. An inference-time projection makes non-negativity and domain mass balance exact.
- We report three physics-validity metrics, Divergence Error (DE), Non-negativity Violation Rate (NVR), and Mass-Conservation Error (MCE), beside SSIM, PSNR, and CRPS. NVR expresses the negative-pixel count of Harder et al. (2023) as a fraction of cells. MCE is the domain-level counterpart of their per-cell mass violation, measured as a relative error of the domain mean. We argue for running NVR and MCE as a reference-free gate. On ERA5 our DE values measure a deformation-like term, as Section 3 explains.
- We report the projection and the penalties separately where the design allows, on one synthetic field and on ERA5. We also compare PC-RF with bicubic interpolation and four reimplemented baselines on one year of ERA5 and on three years (2019–2021). The projection certifies non-negativity and mass balance wherever it is applied. The penalties give no perceptual gain that we can separate from training budget or run-to-run variation, and we report that negative result as such.

2 Diffusion and flow matching

Conditional generative downscaling treats the fine field as a sample from a distribution conditioned on the coarse field. Denoising diffusion models (Ho et al., 2020; Song et al., 2021) corrupt a clean field into Gaussian noise through a forward chain and learn the reverse chain conditioned on a context $\mathbf{c}$,

which costs hundreds to thousands of network evaluations per sample. Rectified flow (Liu et al., 2023; Lipman et al., 2023) replaces that stochastic reverse chain with a deterministic transport along straight paths from noise to data, which we adopt for its sampling efficiency at the high spatial resolutions of downscaling.

Objective: Rectified flow learns a velocity field $\mathbf{v}_\theta$ that transports samples from a source p_0 (Gaussian) to a target p_1 (data) along straight trajectories. Given paired samples $(\mathbf{x}_0, \mathbf{x}_1)$, the linear interpolation $\mathbf{x}_t = (1-t)\mathbf{x}_0 + t\mathbf{x}_1$ induces the objective

$$\mathcal{L}_{\text{RF}} = \mathbb{E}_{t, \mathbf{x}_0, \mathbf{x}_1} \left[\|\mathbf{v}_\theta(\mathbf{x}_t, t, \mathbf{c}) - (\mathbf{x}_1 - \mathbf{x}_0)\|_2^2 \right], \quad (1)$$

where $\mathbf{c}$ is the conditioning context. Here and in the penalties below, the squared norm is averaged over grid cells and channels. Inference integrates the ODE $d\mathbf{x}_t/dt = \mathbf{v}_\theta(\mathbf{x}_t, t, \mathbf{c})$ from $t=0$ to $t=1$ with a forward Euler solver in 20 steps, far fewer evaluations than the stochastic chain.

Problem formulation: Let $\mathbf{x}_c \in \mathbb{R}^{H_c \times W_c \times C}$ be a coarse climate field (ERA5 precipitation and u/v wind at 0.25°), and let $\mathbf{x}_f \in \mathbb{R}^{H_f \times W_f \times C}$ be the fine target on a finer grid over the same box. In our setup the fine grid is a square 128×128 grid, about three times finer than the coarse grid in latitude and about two times finer in longitude. We train $\mathbf{v}_\theta(\mathbf{x}_t, t, \mathbf{c})$ conditioned on $\mathbf{c} = \mathbf{x}_c^\uparrow$, the coarse field bicubically upsampled to the fine grid and stacked with four channels reserved for satellite land-surface context, which are zero in every run we report.

Physics penalties: We augment Equation (1) with three differentiable penalties evaluated on the one-step clean-field estimate $\tilde{\mathbf{x}}_f = \mathbf{x}_t + (1-t)\mathbf{v}_\theta(\mathbf{x}_t, t, \mathbf{c})$. For the horizontal wind $(\tilde{u}, \tilde{v})$ we penalize the squared divergence,

$$\mathcal{L}_{\text{div}} = \mathbb{E} \left[\left\| \frac{\partial \tilde{u}}{\partial x} + \frac{\partial \tilde{v}}{\partial y} \right\|_2^2 \right], \quad (2)$$

with derivatives by central finite differences on the grid. The penalty pulls the wind toward the non-divergent approximation, and it acts on the total divergence, including any divergence the coarse wind already carries. On ERA5 the same operator computed a deformation-like term, for the reason given in Section 3. Precipitation is non-negative, so we penalize $\mathcal{L}_{\text{nn}} = \mathbb{E} \left[\|\min(\tilde{p}, 0)\|_2^2 \right]$. The fine-resolution precipitation mean should match the coarse mean, which keeps the generator from adding or removing precipitation at the domain scale.

$$\mathcal{L}_{\text{mass}} = \mathbb{E} \left[\left| \frac{1}{H_f W_f} \sum_{i,j} \tilde{p}_{ij} - \frac{1}{H_c W_c} \sum_{i',j'} p_{i'j'}^c \right|^2 \right]. \quad (3)$$

The full loss combines the flow objective with the penalties, $\mathcal{L} = \mathcal{L}_{\text{RF}} + \lambda_1 \mathcal{L}_{\text{div}} + \lambda_2 \mathcal{L}_{\text{nn}} + \lambda_3 \mathcal{L}_{\text{mass}}$, with the λ_i held fixed at the values given in Section 3.

Why these constraints: The three penalties encode physics that a downscaled field should respect to be usable downstream. The non-divergent approximation holds for synoptic-scale flow aloft. It is weaker for 10 m wind, where friction, land-sea contrast, and orography make low-level divergence physically real, so the divergence penalty acts as a soft prior and carries no guarantee. Precipitation is a non-negative accumulation. Domain mass balance ties the fine domain mean to the coarse domain mean, a single number that constrains the regional water budget and no individual basin. A field that violates these constraints can still score well on perceptual metrics, so we measure validity directly.

Inference-time projection: The penalties encourage validity but do not enforce it. After integrating the sampling ODE we apply a projection Π to the precipitation channel of the sampled field $\hat{\mathbf{x}}_f$. A rectification clamps precipitation at zero, and a multiplicative rescaling restores the coarse mean exactly,

$$\hat{\mathbf{x}}_f \leftarrow \Pi(\hat{\mathbf{x}}_f; \mathbf{x}_c). \quad (4)$$

Equation (4) makes NVR zero for any model. It also makes MCE zero whenever the rectified field has at least one positive cell, up to floating-point rounding far below the precision we report. The zeros we report show that an all-dry rectified field under a wet coarse mean never occurred. The penalties are meant to shape the field, and the projection guarantees the two hard constraints.

What the projection changes: The projection is two operations, and neither can invent structure. A rectifier sets every negative precipitation cell to zero. It touches only the cells that were negative, and it raises the domain mean because it deletes negative values. The whole channel is then multiplied by one scalar per sample, the ratio of the coarse mean to the mean of the rectified field. A single global factor leaves every ratio between cells unchanged and multiplies every spatial gradient by that same factor, so the rescaling moves the amplitude of the field and leaves its pattern in place. The rectifier is the only step that changes the pattern, and it changes it only where the field was negative. On our data the reference a model learns from is zero over the sea and accumulates over a longer window than the coarse input (Section 3). A trained model therefore need not match the coarse mean, and the factor rescales every cell to close that gap. The unprojected baselines learn the same reference, and their MCE runs from 6.5 to 13.2 in Table 2, so our own scalar may sit far from one. Where a model keeps the sea dry, as the reference does, the rescaled land cells also carry the share of the coarse mean that fell over the sea. We did not save the field before projection, so we did not record two quantities that would complete the answer. They are the fraction of cells the rectifier touches and the distance of the scalar from one. One sampling pass with the projection switched off would give both, and we did not run it. For scale, the baselines we score without the projection reach NVR values from 0.053 to 0.826 in Table 2. None of them carries our non-negativity penalty, so those rates do not bound what our own sampler produces.

Why the wind is not projected: Non-negativity is a pointwise constraint and domain mass is a single scalar, so each has a closed-form projection that costs one pass over the field. A divergence-free projection has no such closed form. It requires solving a Poisson equation for a velocity potential and subtracting its gradient, which is a global operation over the whole grid. Two further reasons explain our choice. Our setup has no fine wind target, so the wind channels carry the upsampled coarse wind, and projecting them would move the field away from the only reference the perceptual scores use. Zero horizontal divergence is also an approximation and not an exact law, because low-level convergence is physically real and mass continuity turns it into ascent. Forcing it to zero would delete signal that belongs in the field. For the same reason the divergence penalty stays soft. On ERA5 it acted on the deformation-like term of Section 3, and Section 4 cannot separate its measured effect there from run-to-run variation. The asymmetry is why we report DE as a diagnostic and treat only non-negativity and mass balance as certificates.

Architecture: The velocity network $\mathbf{v}_\theta$ is a U-Net (Ronneberger et al., 2015) with residual blocks, skip connections, and three resolution levels (128, 64, and 32 cells on a side) above a 16×16 bottleneck. A sinusoidal time embedding enters every residual block, and a FiLM layer (Perez et al., 2018) also modulates the first feature map with it. Multi-head self-attention runs at the 32×32 level and in the bottleneck. The network takes $\mathbf{x}_t$ concatenated with the conditioning image $\mathbf{c}$ at its first convolution. The code reserves the four extra conditioning channels for Sentinel-2 Level-2A land-surface context (Drusch et al., 2012). Every run we report fills them with zeros and trains no land-surface encoder, so no land-surface information reaches the model and we report no Sentinel-2 results. The trained model has 7.6M parameters, small by design, and Figure 1 summarizes the pipeline.

3 Experiments

We ask whether physics constraints reduce physical-law violations without degrading perceptual quality, and we separate the effect of the training penalties from that of the inference projection where the design allows. We report SSIM, PSNR, and probabilistic CRPS, computed over the three channels together, alongside three physics-validity metrics. A sample is one time step over the whole box. For

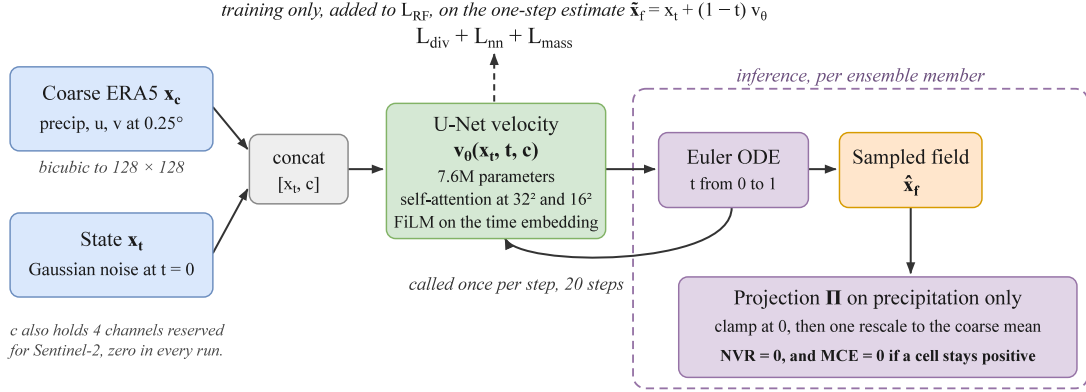


Figure 1: The diagram shows the PC-RF framework as it was trained. The coarse ERA5 field $\mathbf{x}_c$ is bicubically upsampled and concatenated with the state $\mathbf{x}_t$. The U-Net velocity field $\mathbf{v}_\theta(\mathbf{x}_t, t, \mathbf{c})$ uses self-attention only at its two coarsest resolutions and FiLM only for the time embedding. During training the three penalties are added to the flow loss and act on the one-step estimate $\hat{\mathbf{x}}_f$. At inference a forward Euler solver integrates the rectified-flow ODE in 20 steps to the sampled field $\hat{\mathbf{x}}_f$. The projection Π then makes non-negativity and domain mass balance exact for the precipitation channel.

the sampled models (the two flow models, DDPM, and CorrDiff), SSIM, PSNR, and the three validity metrics are computed on the mean of 8 ensemble members, and CRPS on the members themselves. The deterministic baselines are scored on their single output. Averaging smooths a field, so these scores are not strictly comparable across the two groups. The SSIM and PSNR data range comes from the reference in each consecutive test batch of up to 16 samples. Every score is a mean of per-batch means, so the shorter final batch counts as much as a full one. Divergence Error (DE) is the mean $|\partial \hat{u} / \partial x + \partial \hat{v} / \partial y|$ over the test set, taken by central differences in grid units on the standardized wind channels, so it is dimensionless. Precipitation enters the model divided by the standard deviation of the coarse training precipitation, with no mean removed, so zero stays zero and NVR counts truly negative values. SSIM, PSNR, and CRPS are computed in these scaled units, so CRPS carries no physical unit. Non-negativity Violation Rate (NVR) is the fraction of precipitation cells with $\hat{p} < 0$, counted with no tolerance. Mass-Conservation Error (MCE) is the relative error between the domain-mean fine precipitation and the coarse mean, computed per sample and averaged over the test set, so a time step with little coarse precipitation weighs heavily.

Data: The coarse input is ERA5 total precipitation and 10 m u/v wind at 0.25° (Hersbach et al., 2020) over a western European box (40 to 50°N, 10°W to 5°E) at 00, 06, 12, and 18 UTC for 2020. The fine precipitation target is ERA5-Land total precipitation at 0.1° over the same box and hours. We resample it to the square 128 × 128 grid, about 0.08° by 0.12° per cell, and set the ocean cells of that land-only product to zero. ERA5 is the ECMWF fifth-generation reanalysis, which assimilates surface, upper-air, and satellite observations into one global record. ERA5-Land replays the land component at 0.1° under ERA5 forcing (Muñoz-Sabater et al., 2021). Total precipitation is one of the forcing fields it inherits and is not a land-model output. ECMWF states that such fields are purely interpolated onto the finer grid (ECMWF, 2025). Our fine reference therefore carries no precipitation structure that ERA5 does not already hold. The two products also document different accumulation conventions (ECMWF, 2025). An ERA5 value covers the hour ending at the validity time (ECMWF, 2026). An ERA5-Land value accumulates from 00 UTC, so at 06, 12, and 18 UTC it covers 6, 12, or 18 hours, and at 00 UTC it covers the previous 24 hours. The coarse and fine precipitation we compare therefore never cover the same interval, and we do not de-accumulate either field. The mass penalty, the projection, and MCE all use a coarse mean taken over land and ocean alike. Over the sea the zero-filled reference carries

less water than the coarse total, and over land its longer window can make it carry more. A zero MCE therefore certifies agreement with the coarse input alone. We use no fine wind target, so the fine wind channels carry the bicubically upsampled coarse wind. The reference therefore has a nonzero DE of its own, which is the DE of the bicubic row in Table 2 and Table 4. Bicubic reproduces the wind reference exactly, so its gaps to the learned models in SSIM, PSNR, and CRPS come from precipitation and from any wind error of those models.

Divergence on the ERA5 grid: Our ERA5 and ERA5-Land arrays keep the stored latitude order of the source files, so row 0 is the northern edge of the box. The central difference taken down the row index is therefore the negative of the northward derivative. On ERA5 the operator in Equation (2) and in DE thus computes $\partial u/\partial x - \partial v/\partial y$, a stretching-deformation term, and not the divergence. The two wind channels are also standardized separately, and the $10^\circ \times 15^\circ$ box is resampled to a square grid. Even with the correct sign, the two derivative terms would carry unequal weights, set by the two standard deviations and the cell aspect, so the sum would not be proportional to the physical divergence. Every ERA5 DE value and the ERA5 divergence penalty refer to that deformation-like quantity. The synthetic check builds its fields in grid coordinates and is not affected.

Training and baselines: The year gives one sample per time step, split chronologically 80/10/10 into training, validation, and test, so the test set is the final tenth of the year, from 25 November to 31 December 2020. No reported run uses the validation tenth, because no run has a validation loop and every model is scored from its final-epoch checkpoint. The training split holds 1171 samples, which gives 36 optimization steps per epoch at batch 32. The velocity network is the 7.6M-parameter U-Net of Section 2, trained with AdamW for 300 epochs at batch size 32, learning rate 2×10^{-4} , and physics weights $\lambda_1 = \lambda_2 = \lambda_3 = 1.0$ on one NVIDIA A100-40GB. The rectified-flow models are sampled with 20 forward Euler steps and 8 ensemble members. The DDPM and CorrDiff reimplementations use 100 stochastic DDIM steps ($\eta = 1$) and 8 members, and the deterministic baselines give a single member, for which CRPS reduces to the mean absolute error. Bicubic interpolation needs no training. SRCNN (Dong et al., 2016), SRGAN (Ledig et al., 2017), a DDPM downscaler (Ho et al., 2020), and CorrDiff (Mardani et al., 2025) are our reimplementations and not the original authors’ released models. They are trained on the same split and scored on the same held-out test set without the projection. Training budgets differ across arms. PC-RF and the four learned baselines train for 300 epochs. SRCNN, SRGAN, and DDPM use batch 32, and CorrDiff uses batch 16 to fit its two U-Nets in 40 GB, so CorrDiff takes about twice as many optimizer steps. The RF-base row is the no-physics arm of the ablation run (Table 3), trained for 200 epochs at batch 16. It sees fewer samples than PC-RF but takes more optimizer steps (14,600 against 10,800). Every trained ERA5 arm in this paper is a single training run, and neither its training nor its sampling uses a fixed seed.

Synthetic mechanism check: Before reporting on real reanalysis, we isolate the constraint mechanism on one synthetic 128×128 field whose ground truth is physically exact by construction (divergence-free wind from a stream function, non-negative precipitation from Gaussian blobs, a known coarse mass). The field keeps the constraint pathway apart from reanalysis confounds. Table 1 compares a corrupted baseline field with the PC-RF mechanism applied to that field. The baseline is the truth plus spatially correlated noise on all three channels, and its precipitation also carries a constant negative bias of 0.18 times the standard deviation of the true precipitation. The PC-RF arm runs 600 gradient steps on the three penalties, weighted 2, 4, and 6 against a fidelity term of weight 1 that keeps the field near the baseline, and then applies the projection of Equation (4). No network is trained in this check. Relative to the baseline, the penalties lower DE by about 30% ($0.0081 \rightarrow 0.0057$), and the projection cannot account for that change because it never touches the wind. The projection removes all non-negativity violations (NVR $0.236 \rightarrow 0$) and drives MCE to zero. Precipitation fidelity improves as well (correlation $0.956 \rightarrow 0.966$, SSIM $0.518 \rightarrow 0.682$). The PC-RF arm carries the penalties and the projection together. The projection clamps the field and rescales it to the true coarse mean, which removes the net shift in

Table 1: The synthetic mechanism check runs on one 128×128 field in pure NumPy, with a fixed seed and no trained network. The rows are synthetic and are not ERA5 results. RMSE, Corr, and SSIM score the precipitation channel only, and SSIM here uses a 7×7 box window, so its values are not comparable with those of Tables 2 to 4. Bold marks the better of the two method rows, and $\uparrow$ ($\downarrow$) means higher (lower) is better.

Method	DE $\downarrow$	NVR $\downarrow$	MCE $\downarrow$	RMSE $\downarrow$	Corr $\uparrow$	SSIM $\uparrow$
Ground truth	0.0000	0.000	0.0000	0.0000	1.000	1.000
Corrupted field (no physics)	0.0081	0.236	0.1838	0.1397	0.956	0.518
PC-RF mechanism (physics)	0.0057	0.000	0.0000	0.1116	0.966	0.682

the domain mean left by the bias and the clamp. The check therefore cannot say how much of that gain the penalties produced. The synthetic results confirm the constraint pathway on one field, and we keep them distinct from the ERA5 results below.

Real ERA5 downscaling: Table 2 places PC-RF and the unconstrained RF-base beside bicubic interpolation and our reimplemented baselines on the single-year task. No single model wins every column, since CorrDiff takes PSNR (28.38) and CRPS (1.092) while PC-RF takes SSIM (0.914). RF-base and PC-RF carry the inference projection and are the only two models with NVR and MCE of exactly zero, so those two checks hold for them by construction. Every learned baseline without the projection leaves between 5% and 83% of its precipitation cells negative (NVR 0.053 to 0.826, on the ensemble mean for DDPM and CorrDiff) and departs from the coarse mean by an MCE between 6.5 and 13.2. We did not score the reference for MCE, but by design it departs from the coarse mean (Data paragraph), so these values mix model error with that offset. Bicubic is the one exception on mass (MCE 0.0003), since its kernel nearly preserves the coarse mean, though its ringing still drives 10% of cells negative. A negative accumulation is physically impossible, and a hydrological model fed such a field would need it clipped first. At one year PC-RF also scores above RF-base (SSIM 0.912 $\rightarrow$ 0.914, PSNR 26.39 $\rightarrow$ 26.59, CRPS 1.188 $\rightarrow$ 1.149), but the two rows had different training budgets. In the matched-budget ablation of Table 3 the SSIM gap closes (0.912 for full PC-RF against 0.913 for RF-base), while full PC-RF keeps a smaller PSNR and CRPS lead (26.54 against 26.40, and 1.161 against 1.186). Every penalty arm leads RF-base on those two scores at one year. With one run per arm we cannot tell that lead apart from run-to-run variation, and at three years no consistent lead remains (Table 3). One caveat applies to every perceptual column here and in Table 4. The fine reference carries no precipitation structure that ERA5 does not already hold, and it covers a longer accumulation window than our coarse input, so these tables show which models stay physically valid and cannot show which model downscales better.

Three-year ERA5 downscaling: To test the penalties at a matched budget and with more data, we retrained every learned model on ERA5 for 2019 to 2021, which gives about three times the training data. Every learned arm trains for 100 epochs, against 300 and 200 at one year. The two rectified-flow arms and CorrDiff train at batch 16, while SRCNN, SRGAN, and DDPM train at batch 32. The chronological split again holds out the final tenth of the record, from 13 September to 31 December 2021. Table 4 reports the held-out test metrics. PC-RF has a DE of 0.0563, against 0.0567 for bicubic and 0.0570 for RF-base, and that gap is smaller than the spread of the three-year ablation arms (0.0558 to 0.0586, Table 3). RF-base, which shares the projection and trains without penalties, matches PC-RF on every perceptual score (SSIM 0.942 vs. 0.939, PSNR 31.79 vs. 31.81, CRPS 0.772 vs. 0.771), and both keep NVR and MCE at exactly zero. CorrDiff again takes PSNR (32.02) and CRPS (0.761), but its MCE grows to 32.4, and DDPM’s reaches 170.2. We did not score the fine reference itself for MCE. It is zero over the sea and accumulates over a longer window than the coarse input, so a model that reproduced it exactly would not score zero either. The large MCE values therefore mix model error with that offset. At a matched budget no penalty gives a perceptual gain that we can separate from run-to-run variation, and the projection provides the validity guarantee.

Table 2: The table reports real single-year ERA5 downscaling (2020), scored jointly over precipitation and two wind channels whose target is the upsampled coarse wind. Bicubic needs no training. SRCNN, SRGAN, DDPM, and CorrDiff are our reimplementations trained for 300 epochs and scored without the projection. RF-base is the no-physics ablation arm trained for 200 epochs at batch 16, and PC-RF trained for 300 epochs at batch 32. NVR and MCE are exactly zero only for the two models that carry the inference projection, and the bicubic row’s DE is the value of the upsampled wind reference. On ERA5, DE measures the deformation-like term of Section 3 and lower is not necessarily better, so the DE column has no arrow and no bold. Bold marks the best value per column, and $\uparrow$ ($\downarrow$) means higher (lower) is better.

Method	SSIM $\uparrow$	PSNR $\uparrow$	CRPS $\downarrow$	DE	NVR $\downarrow$	MCE $\downarrow$
Bicubic	0.885	26.19	1.356	0.0648	0.101	0.0003
SRCNN	0.833	27.74	1.461	0.0738	0.053	11.1365
SRGAN	0.855	27.03	1.506	0.0700	0.177	13.2239
DDPM	0.656	26.18	1.609	0.0715	0.826	7.5011
RF-base	0.912	26.39	1.188	0.0643	0.000	0.0000
CorrDiff	0.864	28.38	1.092	0.0679	0.451	6.4835
PC-RF	0.914	26.59	1.149	0.0644	0.000	0.0000

Ablation: The ablation in Table 3 removes one physics penalty at a time, at 200 epochs and batch 16 on 2020 and at 100 epochs and batch 16 on 2019 to 2021. The one-year full arm is a separate run from the 300-epoch PC-RF of Table 2. The RF-base arms and the three-year full arm also supply rows of the main tables. NVR and MCE are exactly zero in every row, including RF-base, because the projection enforces them after sampling. At one year, removing the divergence penalty shifts DE the most, from 0.0634 for the full model to 0.0649, in the expected direction. The shift is small, and RF-base, which trains with no penalty at all, sits at 0.0643 between the two. Removing the non-negativity or mass penalty moves SSIM by about 0.003 or less (0.912 to 0.915 across all arms). Every penalty arm leads RF-base on PSNR and CRPS at one year, and no such ordering holds at three years. The three-year block adds an arm that drops the four zero-filled satellite channels from the input. Those channels carry no information, so the arm sees what the full model sees and differs from it only in input width and in the random draws of an unseeded run. Its DE is 0.0586 against 0.0563, and its CRPS is 0.782 against 0.771. That DE gap is larger than the shift from removing the divergence penalty at either scale, so we cannot separate the DE shifts in this table from variation between runs. Removing the mass penalty at three years gives the lowest DE of any arm (0.0558) and an SSIM of 0.942, level with RF-base at the precision we report.

4 What the validity metrics certify, and what they cannot

Two of the three validity metrics are certificates. After the projection of Equation (4), NVR and MCE are zero whatever the model’s training objective, so a zero there reports the inference step and not the learned weights. A certificate cannot rank two models that share it, and Table 4 shows exactly that. PC-RF and RF-base pass through the same projection, both report NVR 0.000 and MCE 0.0000, and their perceptual scores differ by about 0.003 in SSIM, 0.02 dB in PSNR, and 0.001 in CRPS.

DE is a diagnostic, not a certificate. The projection rectifies and rescales precipitation and never touches the wind, so nothing in the method forces the divergence error toward zero. On the synthetic field, where DE measures the true divergence, the penalty lowers it by about 30% (0.0081 to 0.0057). On ERA5 2020 the arm without the divergence penalty sits about 2% above the full model (0.0649 against 0.0634), and RF-base, which has no penalty at all, sits between them at 0.0643. Two three-year arms that differ only in four zero-filled input channels differ by about 4% (0.0586 against 0.0563), so we cannot attribute the ERA5 shift to the penalty. The ERA5 values also measure the deformation-like quantity described in Section 3. The meaning of a DE value depends on the reference. On the synthetic

Table 3: The ablation runs on real ERA5 at 200 epochs and batch 16 on 2020 and at 100 epochs and batch 16 on 2019 to 2021. The divergence, non-negativity, and mass rows each remove one physics penalty. The satellite-channel arm removes the four zero-filled input channels and exists only in the three-year runs. The one-year RF-base checkpoint also supplies the RF-base row of Table 2, and the three-year full and RF-base checkpoints supply the PC-RF and RF-base rows of Table 4, each rescored there with a fresh 8-member sample. NVR and MCE are exactly zero for every arm, so we omit them. DE has no arrow and no bold for the reason given in Table 2. Bold marks the best unrounded SSIM, PSNR, and CRPS value per column within each block.

Configuration	2020 (one year)				2019–2021 (three years)			
	SSIM↑	PSNR↑	CRPS↓	DE	SSIM↑	PSNR↑	CRPS↓	DE
Full PC-RF	0.912	26.54	1.161	0.0634	0.939	31.80	0.771	0.0563
– divergence penalty	0.912	26.58	1.146	0.0649	0.939	31.78	0.773	0.0572
– non-negativity penalty	0.913	26.55	1.160	0.0644	0.940	31.86	0.771	0.0570
– mass penalty	0.915	26.50	1.166	0.0639	0.942	31.80	0.783	0.0558
– satellite channels	n/a	n/a	n/a	n/a	0.939	31.89	0.782	0.0586
RF-base (no physics)	0.913	26.40	1.186	0.0643	0.942	31.79	0.772	0.0570

Table 4: The table reports three-year ERA5 downscaling, trained on 2019 to 2021 and scored jointly over precipitation and two wind channels whose target is the upsampled coarse wind. The final tenth of the record is the test set, and every learned arm trains for 100 epochs. Bicubic needs no training. The learned baselines are our reimplementations scored without the projection, and every arm shares the same split. NVR and MCE are zero only for the two projected models, RF-base and PC-RF. DE has no arrow and no bold for the reason given in Table 2. Bold marks the best value per column, and ↑ (↓) means higher (lower) is better.

Method	SSIM↑	PSNR↑	CRPS↓	DE	NVR↓	MCE↓
Bicubic	0.927	31.48	0.873	0.0567	0.119	0.0006
SRCNN	0.832	30.43	1.057	0.0618	0.032	47.8114
SRGAN	0.869	31.49	0.888	0.0601	0.324	13.7981
DDPM	0.676	28.61	1.173	0.0648	0.054	170.1641
RF-base	0.942	31.79	0.772	0.0570	0.000	0.0000
CorrDiff	0.820	32.02	0.761	0.0598	0.108	32.3586
PC-RF	0.939	31.81	0.771	0.0563	0.000	0.0000

field the true wind is divergence-free, so every unit of DE is spurious. On ERA5 the fine wind reference is the upsampled coarse wind, whose own DE is the bicubic row’s 0.0648 at one year and 0.0567 at three. The flow models of Table 2 and Table 4 sit within 1% of that reference level, and some sit slightly below it. Averaging 8 members smooths the wind, which is one plausible reason for the values below the reference level.

A field can pass the gate and still be wrong, since validity is necessary and not sufficient. Non-negativity fixes a sign at each cell and mass balance fixes one scalar per domain, which leaves nearly every degree of freedom free. We ask a reader to treat NVR and MCE as a gate whose pass condition is the certificate itself, an NVR of zero and an MCE of zero to within floating-point rounding. A pass says that the field is non-negative and matches the coarse domain mean. On our data pair the mass relation the projection enforces is not the physical one, so an MCE of zero here certifies agreement with the coarse ERA5 total and nothing more. We report DE as a diagnostic, and SSIM, PSNR, and CRPS rank the fields that pass the gate. By that condition every unprojected model in Table 2 and Table 4 fails, bicubic included, because we scored them without the projection that any of them could apply. The fine reference would not pass the MCE check either, for the reasons given in Section 3.

The gate is also cheap. NVR, MCE, and DE need only the coarse input and the generated field, and they run in one pass with no simulation. They should therefore run first when verification time is limited. When a field has to be acted on before a reference exists, a check that needs none is the one that can still be run.

Limitations: The most serious limitation is in the data design. Our fine reference carries no precipitation structure that ERA5 does not already hold, since ERA5-Land total precipitation is interpolated from ERA5, the reanalysis that also supplies our coarse input. Its values accumulate from 00 UTC while our coarse input covers only the hour ending at the validity time, and its ocean cells are zero-filled while the coarse mean counts rain over the sea. The reference therefore does not satisfy the coarse-to-fine mass relation that our penalty and our projection target, and we corrected neither mismatch. The null result of Table 4 is consistent with two readings that we cannot separate here. A soft conservation penalty may add nothing once the constraint is enforced at inference. The penalty may also fail to help because this pair of products does not satisfy the relation it enforces. Testing the mass penalty on an independent target, over matched accumulation windows, is the experiment this paper does not contain.

The remaining limitations concern scope and replication. We report one western European domain and one variable set on one target grid, with no evaluation conditioned on extremes, although heavy-rain and strong-wind days are where these checks matter most. Both test sets fall in autumn and early winter. We report no seasonal or regional breakdown and no test of generalization under distribution shift, since the chronological split makes every test set the final stretch of the record. Every arm is one training run with no confidence intervals, so SSIM differences of a few thousandths cannot be separated from run-to-run variation. Training budgets are not matched across arms. At one year, PC-RF and the four learned baselines train for 300 epochs, with CorrDiff at batch 16 and the others at batch 32. The one-year RF-base row and the ablation arms train for 200 epochs at batch 16, and every three-year arm trains for 100 epochs. The comparison across data sizes therefore changes the training budget and the test period along with the data. The baselines are our reimplementations scored without the projection, so whether a projected CorrDiff keeps its PSNR and CRPS lead is untested. On ERA5 the divergence penalty and DE act on the deformation-like quantity of Section 3, so these runs do not measure the physical divergence of the generated wind. No Sentinel-2 input reached any model.

5 Related work

Deep generative models offer several routes from coarse to fine. Convolutional networks (Dong et al., 2016) and then adversarial networks (Goodfellow et al., 2014; Ledig et al., 2017) drove single-image super-resolution. Both were adapted to the geosciences, where stacked super-resolution networks downscale climate projections (Vandal et al., 2017) and adversarial networks recover fine-scale wind and solar structure (Stengel et al., 2020) and downscale time-evolving fields with calibrated spread (Leinonen et al., 2021). Normalizing flows (Rezende and Mohamed, 2015; Dinh et al., 2017; Kingma and Dhariwal, 2018) give exact likelihoods and have been applied to unsupervised climate downscaling (Groenke et al., 2020). Denoising diffusion (Ho et al., 2020; Song et al., 2021) now underlies regional downscalers such as CorrDiff (Mardani et al., 2025) and global ensemble forecasters such as GenCast (Price et al., 2025). Rectified flow (Liu et al., 2023; Lipman et al., 2023) learns a deterministic transport that samples in far fewer steps on image benchmarks, which is why PC-RF builds on it.

A second line of work makes the constraint hard. Harder et al. (2023) add constraint layers to deterministic and generative downscalers that make the fine-grid mean inside every coarse cell equal the coarse value. Their softmax layer also keeps every output non-negative, so both conditions hold exactly in every cell. Most of their layers improve standard fidelity metrics over the same network without them. Their multiplicative layer can apply any function to the predictions before it rescales them, and a positive function guarantees a positive output. They also note that the constraint window can be widened until one window covers the whole domain. Hess et al. (2022) use one global constraint

of this kind in a generative adversarial network that corrects the precipitation of an Earth system model. After training, one factor rescales the output so that its global total matches the model input. The projection in Equation (4) applies the same domain-level rescaling to the rectified field once the ODE has been integrated. A rectifier returns zero for every negative input, so the output is non-negative, and the rescaling needs at least one positive cell. The projection is weaker than the per-cell layers, and it leaves redistribution inside the domain free. [Bochow et al. \(2026\)](#) enforce conservation on the coarse grid at inference in a consistency-model downscaler. Within flow matching, ECI sampling ([Cheng et al., 2025](#)) and PCFM ([Utkarsh et al., 2025](#)) make pretrained models of PDE systems satisfy hard constraints exactly by correcting intermediate states during sampling. [Beuclet et al. \(2021\)](#) compare loss-based and architecture-based conservation constraints in neural emulators of convection and find that the architectural form holds the conservation laws to machine precision without degrading skill. Outside climate, DC3 ([Donti et al., 2021](#)) completes a partial solution to satisfy equality constraints and corrects it with unrolled gradient steps for the inequalities.

Most generative downscalers treat physical consistency as a diagnostic. Such a model can minimize a perceptual or likelihood objective and still miss the fine-scale spatial correlations that make a downscaled field physically plausible. Classical statistical downscaling ([Maraun and Widmann, 2018](#)) captures spatial structure but leans on a stationarity assumption and oversmooths, which washes out the fine-scale detail that drives regional risk. Physics-informed learning ([Raissi et al., 2019](#)) embeds governing equations as soft penalties, and learned solvers and equivariant flows ([Kochkov et al., 2021](#); [Köhler et al., 2020](#)) build physical structure into the architecture. None of these three works addresses conditional downscaling. Geostatistics-informed diffusion ([Ghosh et al., 2024](#)) embeds a spatial prior in a conditional diffusion downscaler for sea level. Adaptive flow matching with soft per-cell conservation penalties ([Debeire et al., 2026](#)) is the closest flow-matching relative of PC-RF. PC-RF differs by making two of its constraints exact with a projection applied after the sampling ODE has been integrated.

6 Conclusion

We introduced PC-RF, a rectified-flow downscaler whose three differentiable penalties on a one-step clean-field estimate target wind divergence, negative precipitation, and mass drift from the coarse integral, and whose inference-time projection makes the last two constraints exact. On ERA5 the divergence operator took the northward derivative with the wrong sign, so those runs penalized and measured a deformation-like term. On one synthetic field the penalties lower DE and the projection removes every NVR and MCE violation, though that arm carries both mechanisms, so the reconstruction gain cannot be split. On the held-out final tenth of 2020, the two projected models are the only ones with NVR and MCE of exactly zero, while our CorrDiff reimplementations wins PSNR and CRPS and leaves 45% of its ensemble-mean precipitation cells negative. On 2019–2021 at a matched budget, RF-base matches PC-RF on every perceptual score and both keep NVR and MCE at exactly zero, so the projection provides the guarantee. Our reanalysis pair cannot settle what the penalties are worth. Its fine reference is interpolated from ERA5 and covers a different accumulation window, so it cannot be compared directly with our coarse input. The projection secures non-negativity and the coarse domain mean at inference time, whether or not the penalties were used. We therefore propose NVR and MCE as a reference-free gate to run before any perceptual score, with a correctly computed DE as its diagnostic. A field that fails the gate needs a correction before downstream use. The projection is one such correction when the coarse and fine fields cover the same area and the same accumulation window.

Future work: The reference comes first. A regional reanalysis that produces its own precipitation, such as CERRA at 5.5 km over Europe ([Ridal et al., 2024](#)), would give a fine target that is not interpolated from ERA5, and its offline surface analysis blends that field with rain-gauge observations. Matching the accumulation windows is part of the same fix. We will also correct the sign of the northward derivative in the divergence operator, penalize divergence relative to the upsampled coarse wind, and retrain. After

that we will score every baseline through the same projection, run every arm over multiple seeds at a matched budget, and extend the study to other regions, seasons, and extreme events. The Sentinel-2 evaluation and a thermodynamic-consistency penalty remain planned.

Broader impact: We target scientific decision support, where negative precipitation or mass drift can mislead hydrological models, a risk the projection reduces. PC-RF was trained on one domain and scored on one held-out period, so it should not drive operational forecasting before broader evaluation across regions, seasons, and extremes.

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